



College Management System

Comprehensive System Manual & Technical Blueprint.

"Workflows, Operational Processes, Data Schemas, and Role-Based Governance for 19 Enterprise Modules."

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MODULE 01

ADMISSION MANAGEMENT

SYSTEM TYPE

100% Online Portal

SEAT ALLOCATION

Cutoff Engine

STUDENT ID

Auto-Generated

1. Key Sub-Modules & Core Capabilities

Online Application Portal | Quota Matrix | Document Verification | Fee Receipt Sync | Seat Allotment Engine | Roll Number Allocation | Branch Placement

2. Workflow & Architectural Data Flow

[Prospect Reg] -> [Document Verify] -> [Cutoff Calculation] -> [Quota Check] -> [Seat Allot & Roll No]

3. Operational Features & Business Rules

- [+] Interactive student application portal with real-time validation checks for document uploads.
- [+] Merit list calculation and automated quota screening (General / Management / NRI).
- [+] Automatic student profile creation and institutional registration number generation upon fee payment.

4. Detailed Working Process & Execution

Prospects register online and upload certificates. System validates cutoff percentages, assigns document verification slots, records quota reservations, and locks seats upon fee payment receipt. Finally, branch allocations and roll numbers generate.

5. Implementation Stack & Progress Blueprint

UI Stack: RegisterStudent.tsx, AdmissionDashboard.tsx
Database: students, departments, admission_applications

Backend: authService.ts, communicationService.ts
Status: Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 02

STUDENT INFORMATION SYSTEM (SIS)

PROFILE RECORD 360 Card View	CERTIFICATES Instant Generator	ALUMNI BRIDGE Auto-Graduation
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1. Key Sub-Modules & Core Capabilities

Student 360 Card | Academic History | Certificate Generator | Parent Contact Sync | ID Card Printer | Discipline Register | Graduation Audit

2. Workflow & Architectural Data Flow

[Enrolled Student] -> [360 Master Profile] -> [Conduct History] -> [Certificate Generator] -> [Alumni Sync]

3. Operational Features & Business Rules

- [+] Central 360-degree digital master profile card for every enrolled student.
- [+] Automated certificate generator (Bonafide, Transfer, Conduct, Course Completion).
- [+] Continuous record tracking that seamlessly archives profiles into the Alumni database upon graduation.

4. Detailed Working Process & Execution

Students input profile details and upload photo proofs. The system formats printable digital ID cards, maps academic semester records, tracks disciplinary logs, and updates parent details. On graduation, profiles auto-transfer to the Alumni database.

5. Implementation Stack & Progress Blueprint

UI Stack:	StudentDashboard.tsx, StudentProfile.tsx	Backend:	studentService.ts
Database:	students, certificates, parent_details	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 03

ACADEMIC MANAGEMENT

CURRICULUM
CBCS Regulation

ELECTIVES
Merit Auto-Alloc

TIMETABLE
Clash-Free Engine

1. Key Sub-Modules & Core Capabilities

Academic Calendar | Semester Structuring | Course Registration | Curriculum Regulations | Credit Banking | Open Electives | Visual Timetables

2. Workflow & Architectural Data Flow

[Regulation Setup] -> [Curriculum & Semester] -> [Clash-Free Timetable] -> [Elective Selection] -> [Enrollment Approved]

3. Operational Features & Business Rules

- [+] Configurable curriculum regulations (Choice Based Credit System - CBCS / R23 / R20).
- [+] Automatic elective course allocation engine based on student preference and merit ranks.
- [+] Clash-free visual timetabling matrix mapping classroom capacity and faculty availability.

4. Detailed Working Process & Execution

Deans create semesters and define course codes under regulations. Faculty submit weekly lecture availability. The system compiles visual timetables. Students log in during course registration windows to select open electives.

5. Implementation Stack & Progress Blueprint

UI Stack:	DeanAcademicAdmin.tsx, GovernanceView.tsx	Backend:	academicService.ts
Database:	curriculums, courses, semesters, timetables	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 04

ATTENDANCE

TRACKING MODE
Biometric / RFID

SHORTAGE ALERT
75% Threshold

PARENT ALERT
Automated SMS

1. Key Sub-Modules & Core Capabilities

Biometric RFID Sync | Daily Class Register | 75% Shortage Tracker | Parent SMS Trigger | Duty Leave Approvals | Department Analytics

2. Workflow & Architectural Data Flow

[RFID Hardware Check-in] -> [Class Register Entry] -> [75% Shortage Run] -> [Warning List] -> [Parent SMS Dispatch]

3. Operational Features & Business Rules

- [+] Real-time biometric RFID hardware integration pulling turnstile punch logs directly to DB.
- [+] Automated attendance shortage calculation flags students below the mandatory 75% threshold.
- [+] SMS messaging engine dispatching immediate absentee alerts to registered parent mobile numbers.

4. Detailed Working Process & Execution

Biometric devices capture entry logs. Faculty review and verify class sheets. The shortage calculator evaluates weekly compliance scores, generates warning letters, and triggers SMS alerts to parents.

5. Implementation Stack & Progress Blueprint

UI Stack:	AttendanceModule.tsx, AttendanceNotifications.tsx	Backend:	attendanceService.ts, notificationController.js
Database:	attendance_logs, attendance_shortages, notification_logs	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 05

EXAMINATION (CONTROLLER SUITE)

EVALUATION
SGPA/CGPA Engine

SECURITY
AES-256 Lock

REVALUATION
Instant Grade Sync

1. Key Sub-Modules & Core Capabilities

Exam Schedules | Hall Ticket Clearance | Encrypted Question Bank | Exam Centers | Invigilator Roster | Seating Matrix | Grace Marks Board | Malpractice Verdicts

2. Workflow & Architectural Data Flow

[Exam Schedule & Encrypt QP] -> [Hall Ticket Clearance] -> [Seating Matrix] -> [Moderation & Grace Marks] -> [COE Result Publish]

3. Operational Features & Business Rules

- [+] Dedicated Controller of Examinations (COE) Governance Suite with outcome-based SGPA/CGPA computation.
- [+] Automated Hall Ticket clearance and withhold controls for fee defaulters or low attendance (<65%).
- [+] AES-256 Question Paper encryption, invigilation duty rosters, and anti-malpractice seating matrix.
- [+] Board moderation, revaluation grade updates, grace marks (+1/+2) sanctioning, and malpractice verdicts.

4. Detailed Working Process & Execution

Exam branch creates timetables and allocates centers. Candidates clear fee/attendance checks to unlock hall tickets. Question papers undergo AES-256 encryption. Hall invigilators track live attendance. Internal and external marks are moderated, results process to SGPA/CGPA, and grade cards publish.

5. Implementation Stack & Progress Blueprint

UI Stack:	DeanExaminationAdmin.tsx, ExaminationControlModule.tsx	Backend:	examService.ts
Database:	exam_schedules, hall_tickets, results, malpractice_cases	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 06

FACULTY ERP

WORKLOAD
Weekly Tracker

RESEARCH LOG
Scopus & NAAC

APPRAISAL
Self-Assessment

1. Key Sub-Modules & Core Capabilities

Faculty Profile | Workload Manager | Leave Portal | Attendance Register | Research Publications | Grant Tracker | Performance Appraisal | Payroll Ledger

2. Workflow & Architectural Data Flow

[Workload Allocation] -> [Lecture & Lab Run] -> [Research Pub Log] -> [Appraisal Scoring] -> [Payroll Review]

3. Operational Features & Business Rules

- [+] Workload manager tracking weekly lecture, tutorial, and lab hours against statutory guidelines.
- [+] Research and publication repository cataloging Scopus, Web of Science, and UGC journal papers.
- [+] Self-appraisal digital sheets linked directly to NAAC/NBA accreditation criteria.

4. Detailed Working Process & Execution

Faculty update profiles, upload research citations, submit leave requests, and track weekly lecture schedules. Department heads review performance ratings, calculate appraisal scores, and approve leave allocations.

5. Implementation Stack & Progress Blueprint

UI Stack:

FacultyDashboard.tsx, FacultyProfile.tsx

Database:

faculty_profiles, workloads, leaves, research_publications

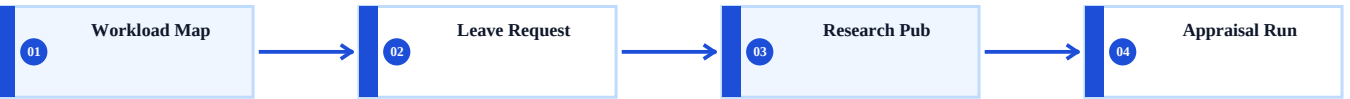
Backend:

facultyService.ts

Status:

Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 07

LEARNING MANAGEMENT SYSTEM

CONTENT HUB

PDF / MP4 / Docs

ASSESSMENT

Online Quiz Key

SUBMISSION

Plagiarism Check

1. Key Sub-Modules & Core Capabilities

Study Materials | Video Lectures | Digital Assignments | Online Quiz Engine | Discussion Forum | Virtual Classroom | Gradebook Sync

2. Workflow & Architectural Data Flow

[Content Upload] -> [Assignment Posting] -> [Timed Online Quiz] -> [Plagiarism Check] -> [Gradebook Auto-Sync]

3. Operational Features & Business Rules

- [+] Central study material repository supporting PDF notes, video lectures, and presentation decks.
- [+] Automated online quiz modules featuring randomized question banks and instant key grading.
- [+] Assignment submission portal with deadline enforcement and plagiarism check indicators.

4. Detailed Working Process & Execution

Faculty create course modules, upload study files, and schedule timed quizzes. Students access study materials, watch video lectures, submit assignments, and participate in discussion threads. Quiz scores auto-sync to gradebooks.

5. Implementation Stack & Progress Blueprint

UI Stack:

LMSCourseView.tsx, AssignmentSubmission.tsx

Database:

lms_courses, materials, assignments, quizzes

Backend:

lmsService.ts

Status:

Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 08

PLACEMENT CELL

ELIGIBILITY

CGPA / Backlog Filter

RECRUITER PORTAL

Job & Drive Post

ANALYTICS

Package & Highest

1. Key Sub-Modules & Core Capabilities

Company Registry | Job Drive Postings | Eligibility Screener | Student Applications | Aptitude Testing | Interview Schedule | Offer Letter Vault | Placement Analytics

2. Workflow & Architectural Data Flow

[Recruiter Drive Post] -> [CGPA & Backlog Filter] -> [Aptitude Test & Interview] -> [Digital Offer Letter] -> [Package Stats]

3. Operational Features & Business Rules

- [+] Automated eligibility criteria filter screening candidates by minimum CGPA and active backlog limits.
- [+] Corporate recruiter portal for posting vacancies, filtering candidate resumes, and managing drive schedules.
- [+] Placement analytics dashboard visualizing branch-wise placement percentages, average packages, and top recruiters.

4. Detailed Working Process & Execution

Placement officer registers corporate recruiters. Companies post vacancies with criteria. Eligible students apply. System schedules online aptitude tests and interview panels. Selected candidates receive digital offer letters.

5. Implementation Stack & Progress Blueprint

UI Stack: PlacementDashboard.tsx, DriveDetailsModal.tsx
Database: placement_drives, job_applications, offer_letters

Backend: placementService.ts
Status: Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 09

HOSTEL MANAGEMENT

ROOM MATRIX
Live Vacancy Map

ATTENDANCE
Night Check-in

PASS SYSTEM
Digital Outing

1. Key Sub-Modules & Core Capabilities

Room Allocation | Night Attendance | Hostel Fee Ledger | Visitor Registry | Outing Gatepasses | Maintenance Complaints | Mess Menu Manager

2. Workflow & Architectural Data Flow

[Online Room App] -> [Fee Verification] -> [Bed Matrix Alloc] -> [Night Biometric Sync] -> [Outing & Complaints]

3. Operational Features & Business Rules

- [+] Live bed matrix displaying vacant, occupied, and maintenance-blocked room allocations in real time.
- [+] Digital night attendance register with mobile biometric check-in integration for wardens.
- [+] Automated complaint ticketing system routing plumbing, electrical, and mess issues to maintenance staff.

4. Detailed Working Process & Execution

Students apply for hostel rooms online. System verifies fee clearance and allocates bed numbers. Wardens record daily night attendance. Outing gatepasses require warden approval and security gate validation.

5. Implementation Stack & Progress Blueprint

UI Stack:	HostelDashboard.tsx, RoomMatrixView.tsx	Backend:	hostelService.ts
Database:	hostel_rooms, hostel_allocations, night_attendance, outing_passes	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 10

TRANSPORT MANAGEMENT

TRACKING Live GPS Coordinates	ROUTES Stop & Capacity Map	PASS CONTROL QR / RFID Pass
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1. Key Sub-Modules & Core Capabilities

Route Management | Bus Capacity Map | Live GPS Tracking | Driver Allocations | Transport Fee Ledger | Digital Bus Passes | Vehicle Maintenance

2. Workflow & Architectural Data Flow

[Route & Stop Setup] -> [Route Opt & Fee Pay] -> [Digital Bus Pass] -> [Live GPS Streaming] -> [Fleet Maintenance]

3. Operational Features & Business Rules

- [+] Route mapper tracking bus stops, fare zones, and seating capacities across fleet vehicles.
- [+] Live GPS vehicle location integration feeding real-time coordinates to student/parent dashboards.
- [+] Digital transport fee collection register linked directly to automated bus boarding pass issuance.

4. Detailed Working Process & Execution

Transport manager defines bus routes, pick-up stops, and assigns drivers. Students select routes and pay transport fees online. Digital bus passes generate, and live GPS coordinates stream during transit hours.

5. Implementation Stack & Progress Blueprint

UI Stack:	TransportDashboard.tsx, GPSTracker.tsx	Backend:	transportService.ts
Database:	bus_routes, vehicles, transport_allocations, bus_passes	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 11

LIBRARY MANAGEMENT

SEARCH ENGINE OPAC Catalog	TRACKING Barcode / RFID	FINE CALCULATOR Auto-Dues Sync
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1. Key Sub-Modules & Core Capabilities

OPAC Book Catalog Barcode Issue / Return Overdue Fine Calculator Digital E-Books Journal Subscription Member Registry Book Reservation
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2. Workflow & Architectural Data Flow

[OPAC Book Catalog] -> [Barcode/RFID Checkout] -> [Return Due Tracking] -> [Overdue Fine Run] -> [Student Fee Dues Sync]
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3. Operational Features & Business Rules

- [+] Online Public Access Catalog (OPAC) index for physical books, journals, and digital research papers.
- [+] Barcode/RFID scanner integration executing check-outs and returns within seconds.
- [+] Automated library fine calculation engine linked directly to student financial fee ledgers.

4. Detailed Working Process & Execution

Librarians scan book barcodes for check-outs. System monitors return due dates and sends overdue reminders. Overdue fines auto-calculate daily and sync to the student fee ledger.

5. Implementation Stack & Progress Blueprint

UI Stack:	LibraryDashboard.tsx, OPACCatalogView.tsx	Backend:	libraryService.ts
Database:	library_books, book_issues, fines, digital_resources	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 12

FINANCE & ACCOUNTING

LEDGER Tuition & Fee Dues	SCHOLARSHIPS Scheme Adjustments	AUDITING GST & Balance Sheet
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1. Key Sub-Modules & Core Capabilities

Tuition Fee Ledger | Online Payment Gateway | Scholarship Adjustments | Staff Payroll Disbursements | Vendor Invoices | GST Compliance | Financial Balance Sheets

2. Workflow & Architectural Data Flow

[Fee Structure Setup] -> [Online Payment Gateway] -> [Scholarship Ledger Adjust] -> [Vendor & Payroll] -> [Annual Audit Sheet]

3. Operational Features & Business Rules

- [+] Tuition fee ledger managing online payment gateways, installment plans, and transaction receipts.
- [+] Scholarship adjustment engine mapping government schemes, merit concessions, and fee waivers.
- [+] Automated balance sheet, P&L statement, TDS filing, and GST compliance report generator.

4. Detailed Working Process & Execution

Finance department generates fee structures. Students pay fees online via payment gateways. Accounts verify scholarship grants, process vendor bills, execute staff payroll disbursements, and generate annual audit ledgers.

5. Implementation Stack & Progress Blueprint

UI Stack:	FinanceDashboard.tsx, FeeReceiptModal.tsx	Backend:	financeService.ts
Database:	fee_structures, fee_payments, scholarships, vendor_invoices	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 13

HRMS & STAFF PAYROLL

PERSONNEL
Digital Staff Files

ATTENDANCE
Biometric Sync

PAYROLL
Scale & Allowances

1. Key Sub-Modules & Core Capabilities

Employee Directory | Recruitment Portal | Biometric Attendance Sync | Leave Approval Workflow | Salary Scale Engine | Pay Slip Generator | Service History Log

2. Workflow & Architectural Data Flow

[Employee Registration] -> [Biometric Punch Sync] -> [Leave Deduction Run] -> [Payroll Calculation] -> [Digital Payslip Release]

3. Operational Features & Business Rules

- [+] Central employee digital repository storing qualifications, service history, and statutory documents.
- [+] Biometric RFID check-in sync evaluating working hours, overtime, and leave deductions automatically.
- [+] Automated payroll disbursement engine calculating basic pay, DA, HRA, PF, TDS, and generating PDF payslips.

4. Detailed Working Process & Execution

HR creates employee profiles. Biometric hardware syncs daily staff attendance. Department heads approve leaves. Payroll engine computes monthly salaries, applies tax/PF deductions, and generates digital payslips.

5. Implementation Stack & Progress Blueprint

UI Stack:	HRMSDashboard.tsx, PayslipViewerModal.tsx	Backend:	hrmsService.ts
Database:	employees, staff_attendance, leave_requests, payrolls	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 14

INVENTORY & ASSET CONTROL

ASSET LEDGER
WDV Depreciation

STOCK ALERT
Low Volume Warning

SECURITY
Gatepass Audit

1. Key Sub-Modules & Core Capabilities

Asset Master Register | Lab Equipment Log | Purchase Order (PO) | Consumables Stock | Depreciation Calculator | Maintenance Tickets | Outward Gatepasses

2. Workflow & Architectural Data Flow

[PO Request & Approve] -> [Asset Log & WDV Depreciation] -> [Stock Level Warning] -> [Maintenance Ticket] -> [Outward Gatepass]

3. Operational Features & Business Rules

- [+] Asset register tracking initial capital costs, location assignments, and Written Down Value (WDV) depreciation.
- [+] Stock consumable indicator raising automatic warnings when lab supplies drop below safety thresholds.
- [+] Outward gatepass log tracking equipment sent out for servicing, calibration, or inter-campus transfer.

4. Detailed Working Process & Execution

Department custodians log equipment details. Purchase requests are approved. Stock levels auto-adjust on item issuance. Maintenance tickets trigger repair schedules, and gatepasses authorize security checks.

5. Implementation Stack & Progress Blueprint

UI Stack:

Database:

InventoryDashboard.tsx, GatepassModal.tsx
assets, purchase_orders, inventory_stock, gatepasses

Backend:

Status:

inventoryService.ts
Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 15

ACCREDITATION (NBA / NAAC)

NAAC SSR
Criteria 1-7 Vault

NBA OBE
CO-PO Attainment

CERTIFICATES
Auto Generator

1. Key Sub-Modules & Core Capabilities

NAAC Criteria Repository | NBA CO-PO Attainment | NIRF / AISHE Exporter | Event Proposals | Participant Check-in | Evidence Vault | Digital Certificate Signer

2. Workflow & Architectural Data Flow

[Event Proposal] -> [Principal & HOD Sign] -> [Event Check-in & Evidence Ingest] -> [CO-PO Attainment] -> [NAAC Vault & Certs]

3. Operational Features & Business Rules

- [+] NAAC Self-Study Report (SSR) evidence repository supporting criteria-wise document categorization.
- [+] NBA Outcome-Based Education (OBE) matrix mapping Course Outcomes (CO) to Program Outcomes (PO).
- [+] Event workflow engine managing approval flows, attendance verification, and auto-signed digital certificates.

4. Detailed Working Process & Execution

Faculty submit event proposals. HOD and Principal approve. Students register and attend. Event coordinators upload reports and photo proofs into the accreditation vault. System generates signed participant certificates.

5. Implementation Stack & Progress Blueprint

UI Stack:	AdminAccreditation.tsx, EventProposalModal.tsx	Backend:	accreditationService.ts, accreditationController.js
Database:	accreditation_events, accreditation_evidence, co_po_mappings	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 16

COMMUNICATION & CIRCULARS

CHANNELS

SMS / Email / Push

BROADCASTING

Instant Dispatch

ARCHIVE

Public Circulars

1. Key Sub-Modules & Core Capabilities

Multi-Channel Broadcast | Official Circulars | Parent SMS Engine | Email Newsletter | Push Notifications | Notice Board Portal | Delivery Logs

2. Workflow & Architectural Data Flow

[Circular Draft] -> [Select Target Roles] -> [Multi-Channel Broadcast] -> [Notice Board Feed] -> [Delivery Audit Log]

3. Operational Features & Business Rules

- [+] Multi-channel broadcast engine dispatching announcements via SMS, Email, Push Notifications, and WhatsApp.
- [+] Official circular archive displaying verified administrative orders on role dashboards.
- [+] Automated trigger notifications for fee payment reminders, exam dates, emergency holidays, and event updates.

4. Detailed Working Process & Execution

Authorized staff draft announcements and select target audiences (Students / Parents / Staff). Dispatch engine transmits bulk messages across channels while simultaneously publishing circulars to user dashboards.

5. Implementation Stack & Progress Blueprint

UI Stack:

AdminCommunication.tsx, BroadcastComposer.tsx

Backend:

communicationService.ts, communicationController.js

Database:

communication_broadcasts, circulars, notification_logs

Status:

Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 17

GRIEVANCE & REDRESSAL

PRIVACY
100% Anonymous Mode

COMPLIANCE
Anti-Ragging Vault

TRACKING
Resolution Tickets

1. Key Sub-Modules & Core Capabilities

Anonymous Complaint Box | Anti-Ragging Committee | Women Cell Portal | Case Assignment | Investigation Log | Hearing Schedule | Resolution Audit

2. Workflow & Architectural Data Flow

[File Complaint (Anonymous/Named)] -> [Committee Ticket Route] -> [Hearing & Evidence] -> [Resolution Order] -> [Case Closed]

3. Operational Features & Business Rules

- [+] 100% anonymous complaint submission option preserving student and staff privacy during sensitive filings.
- [+] Legal compliance trackers for Anti-Ragging, Internal Complaints Committee (ICC), and Women Cell regulations.
- [+] Resolution ticketing system tracking case assignments, hearing dates, investigation notes, and final orders.

4. Detailed Working Process & Execution

Complainant registers a grievance online. System assigns a confidential tracking ticket. Designated committee reviews details, conducts hearings, logs investigation notes, and issues official resolution orders.

5. Implementation Stack & Progress Blueprint

UI Stack:	GrievancePortal.tsx, CaseDetailsModal.tsx	Backend:	grievanceService.ts
Database:	grievances, grievance_hearings, resolution_orders	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 18

ALUMNI RELATIONS

DIRECTORY

Verified Alumni Log

MENTORSHIP

Student Match

DONATIONS

80G Tax Receipt

1. Key Sub-Modules & Core Capabilities

Alumni Directory | Career Job Portal | Mentorship Network | Reunion Events | Donation Gateway | Success Stories | 80G Tax Receipts

2. Workflow & Architectural Data Flow

[Graduation Database Sync] -> [Alumni Directory Update] -> [Mentorship & Job Posting] -> [Donation] -> [80G Tax Receipt]

3. Operational Features & Business Rules

- [+] Verified alumni directory automatically populated from student graduation database records.
- [+] Mentorship network pairing experienced alumni professionals with current students for career guidance.
- [+] Secure donation gateway issuing official tax benefit receipts (80G) for institutional contributions.

4. Detailed Working Process & Execution

Graduating students transition to the alumni directory. Alumni update employment details, post job referral openings, offer mentorship slots, and make institutional donations. System issues automated tax receipts.

5. Implementation Stack & Progress Blueprint

UI Stack:	AlumniDashboard.tsx, MentorshipModal.tsx	Backend:	alumniService.ts
Database:	alumni_profiles, mentorship_requests, alumni_donations	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart



MODULE 19

ADMINISTRATION & SECURITY

RBAC
19-Module Matrix

AUDIT LOG
Immutable Tracking

DB BACKUP
Automated Schedules

1. Key Sub-Modules & Core Capabilities

Role Permissions Matrix | Unified User Control | Immutable Audit Logs | System Configuration | Security Hardening | Automated DB Backups | Global Analytics

2. Workflow & Architectural Data Flow

[Define RBAC Permissions] -> [User Access Hardening] -> [Immutable Audit Trail Log] -> [Encrypted Nightly DB Backup]

3. Operational Features & Business Rules

- [+] Granular Role-Based Access Control (RBAC) engine configuring permissions across all 19 system modules.
- [+] Immutable audit trail logging every user login, database modification, API request, and export action.
- [+] Automated database backup cron-jobs creating daily encrypted snapshots with single-click restore capabilities.

4. Detailed Working Process & Execution

Super admins configure role permissions and assign user credentials. The system logs all user actions in an immutable audit ledger. Nightly cron-jobs create encrypted database backups and monitor system health.

5. Implementation Stack & Progress Blueprint

UI Stack:	AdminGovernanceView.tsx, UserManagementModal.tsx	Backend:	adminService.ts, authMiddleware.js
Database:	users, roles, permissions, audit_logs, backup_logs	Status:	Production-Ready (100% Operational)

6. System Operational Workflow Flowchart

